

Use Case Specifications

Elemental Quest

Cmpt 276 - Group 12

Use Case 1: Start Game

Primary Actor: Player

Goal: To initialize and begin a new level of Elemental Quest with both Ember and Aqua placed on the map.

Preconditions:

- The application is launched on the player's device.
- The main menu is displayed.
- Level data is available and successfully loaded.

Main Scenario (Tasks & Functions):

- The player opens the application.
- The START menu is displayed on the screen.
- The player clicks "Start Game" to begin.
- The system transitions from the START menu to the game map.
- The system loads the selected level, including:
 - Start points for Ember and Aqua
 - Fire Gate and Water Gate
 - Elemental Crystals
 - Golden Gems
 - Shadow Sentinels
 - Hazards (lava and water)
- The score is initialized to 0.
- The timer begins counting.
- The system displays:

- The map
 - Current score
 - Remaining time
 - Life hearts
 - Active character indicator
- Gameplay begins.

Extensions (Variations/Exceptions):

- If level data fails to load, the system displays an error message and returns to the main menu.

System Information & Post-conditions:

- The game is in an active state.
- Both characters are placed at their starting positions.
- The player can move and use abilities.

Use Case 2: Move Character

Primary Actor: Player

Goal: To move the currently controlled character (Ember or Aqua) across the board.

Preconditions:

- The game is active.
- The selected character is alive.

Main Scenario (Tasks & Functions):

- The player presses a movement key (W, A, S, or D).
- The system checks whether the target cell is valid.
- The system verifies the cell is not blocked by a wall or barrier.
- If valid, the character moves one cell in the selected direction.
- The system updates the character's position on the map.
- Shadow Sentinels move accordingly.
- The screen refreshes to display updated positions.

Extensions (Variations/Exceptions):

- **Invalid Move:** If the target cell contains a wall or barrier, the character remains stationary while the game tick proceeds.
- Enemies still move during that tick.

System Information & Post-conditions:

- The character's position is updated or remains unchanged.
- The game continues.

Use Case 3: Collect Elemental Crystal

Primary Actor: Player

Goal: To collect all Elemental Crystals required to unlock the exit gates.

Preconditions:

- Either Ember or Aqua enters a cell containing a crystal.

Main Scenario (Tasks & Functions):

- The system detects that the character has entered a crystal cell.
- The crystal is removed from the map.
- The player's score increases.
- The crystal counter updates.
- If all crystals are collected, the exit gates unlock.

Extensions (Variations/Exceptions):

- If a crystal was already collected, no additional points are awarded.

System Information & Post-conditions:

- The specific crystal is permanently removed from the board.
- The player's score is updated and displayed.

Use Case 4: Elemental Hazard Interaction

Primary Actor: Player

Goal: To process safe or harmful interactions between characters and elemental hazards.

Preconditions:

- A character enters a hazard cell (lava or water).

Main Scenario (Tasks & Functions):

- The system detects the hazard type.

Safe Interaction:

- Ember enters lava → No damage.
- Aqua enters water → No damage.

Harmful Interaction:

- Ember enters water, or Aqua enters lava.
- The system detects harmful interaction.
- The character loses one life heart.
- The life counter updates on screen.

Extensions (Variations/Exceptions):

- If the character loses their final life heart, the system triggers a Game Over state.

System Information & Post-conditions:

- Life hearts are reduced if harmed.
- The level continues until no lives remain.

Use Case 5: Collision with Enemy

Primary Actor: Enemy (Shadow Sentinel)

Goal: To determine the outcome when a Shadow Sentinel collides with a character.

Preconditions:

- A Shadow Sentinel and a character occupy the same cell.

Main Scenario (Tasks & Functions):

- The system detects the collision.
- The character loses one life heart.
- The life counter updates.

Extensions (Variations/Exceptions):

- If it were the character's final life heart:
 - The system stops gameplay.
 - A "Game Over" screen is displayed.

System Information & Post-conditions:

- Life hearts decrease.
- The level ends if no lives remain.

Use Case 6: Level Completion

Primary Actor: Player

Goal: Both characters reach their respective exit gates after collecting all crystals.

Preconditions:

- All Elemental Crystals are collected.
- Ember reaches the Fire Gate.
- Aqua reaches the Water Gate.

Main Scenario (Tasks & Functions):

- Ember moves into the Fire Gate.
- Aqua moves into the Water Gate.
- The system verifies that all crystals are collected.
- The timer stops.
- Final score and time are displayed.
- A “Level Complete” screen appears.

Extensions (Variations/Exceptions):

- **Only One Character at Exit:**
 - If only one character reaches their gate, the level does not complete.
 - The game continues until both are at their gates.

System Information & Post-conditions:

- The level is marked as completed.
- The player may proceed to the next level.

Use Case 7: Loose by Timeout

Primary Actor: System Timer

Goal: To end the level when the time limit expires.

Preconditions:

- The timer reaches zero.

Main Scenario (Tasks & Functions):

- The timer reaches zero.
- The system stops all movement.
- A "Time Over" screen appears.
- The final score is displayed.

System Information & Post-conditions:

- The game session ends.
- The player may restart or return to the menu.